

Mechanism Analysis - 28 Nov 2025



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Learning Objects

1. **What is mechanism analysis?**
2. **When do we need mechanism analysis?**
3. **Different methods**
4. **How to find potential mechanisms?**
5. **Real examples**

Why mechanism analysis ?

For innovations to generate sustained growth:

- ✓ Not only need to know that something works,
- ✓ but also need scientific **explanations for why**

- Joel Mokyr, 2025 Nobel Prize in economic winner

Something works: X to Y; and Causal identification.

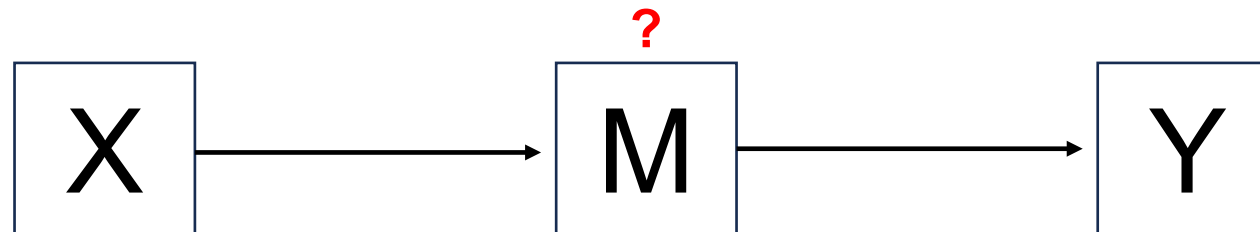
Why? : **Mechanism analysis**

1. What is mechanism analysis?

1. Mechanism /Channel analysis

Definitions:

- Firstly baseline: X affect Y (Regress Y on X with a significant coefficient.)
- Is it causal: X to Y? (Causal inference, econometric causal: IV, DiD, RDD, etc.)
- **Then, How can X affect Y ? or through which channel/mechanism X affects Y?**
- The **most important part** of “empirical story telling”.
- An simple example: football games (X) => - stock return volatility (Y), How?



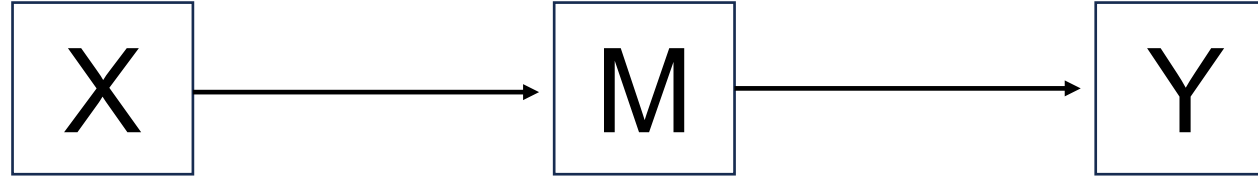
2. When Mechanism /Channel ?

1. No need M: X to Y is so direct: e.g., tax cuts and corporate investments
2. No need M: The X to Y is a direct theory prediction:
 - own paper theory model, or
 - other theory papers

⇒ But: can still do it;

When: Interesting X to Y, but why?

3. Mechanism /Channel analysis: methods



Baseline: $Y = \beta_0 + \beta_1 X + u$

Method 1: X to M and M to Y: $M = \beta_0 + \alpha_1 X + u$; $Y = \beta_0 + \gamma_1 M + u$

Method 2: X to M and “M to Y”: $M = \beta_0 + \alpha_1 X + u$; adding paper supports $Y = \beta_0 + \gamma_1 M + u$

Method 3: X,M to Y: $Y = \beta_0 + \alpha_1 X + \alpha_2 M + u$

Method 4: X as IV of M to Y: $Y = \beta_0 + \gamma_1 \hat{M} + u$; $M = \beta_0 + \alpha_1 X + u$, 2SLS framework

Method 5: $X * M$ to Y: $Y = \beta_0 + \alpha_1 X * M + u$

Method 6: X to Y, M subgroups: $Y = \beta_0 + \beta_1 X + u$.

- 1/2: (Chen et al., 2020, AER, send-down movement);
- 3: (Persico et al., 2004, JPE, height wage premium);
- 4: (Di Giuli and Laux, 2022, JFE);
- 5/6: (Li et al., 2025, CAR; Ye et al., 2023, JAE)

4. How to find potential mechanisms?

What can be M?

Must be logical: X to M to Y

Must be supported by literature: both X to M and M to Y

1. X to M; & M to Y
 2. X to Y different by M
- Literature will provides you: the mechanism and the proxies of the mechanism (M)

5. Real examples

5. Real examples

(Chen et al., 2020, AER): M on X

- The send-down movement: 16 million urban youth mandated to countryside
- X: propensity of urban youth in each country, a DiD: +
- Y: rural education: relevant rural child education

Mechanism: How urban youth + education: + school teachers (M)

5. Real examples: (Chen et al., 2020, AER): Y on X

Dependent variables:	Years of education		Complete primary		Complete junior high	
	Rural	Urban	Rural	Urban	Rural	Urban
Sample:	(1)	(2)	(3)	(4)	(5)	(6)
X	3.237 (0.701)	0.151 (0.517)	0.441 (0.0873)	-0.0658 (0.0611)	0.767 (0.121)	-0.0517 (0.103)

$Y = \beta_0 + \beta_1 X + u$

Dependent variables:	Density of primary school teachers			Density of secondary school teachers		
	All	State-funded	Non-state-funded	All	State-funded	Non-state-funded
School types:	(1)	(2)	(3)	(4)	(5)	(6)
X	0.034 (0.012)	0.005 (0.011)	0.051 (0.028)	0.017 (0.005)	0.010 (0.004)	0.035 (0.007)

$M = \beta_0 + \alpha_1 X + u$

5. Real examples

(Persico et al., 2004, JPE): Y on X,M

- The height wage premium
- X: workers height
- Y: workers hourly wage

Mechanism: Adolescent period height

5. Real examples: (Persico et al., 2004, JPE): Y on X,M

TABLE 3

OLS ESTIMATES ln(Wage) EQUATION FOR ADULT, WHITE MALE WORKERS, NCDS AND NLSY

COVARIATE	BRITAIN: NCDS (N=1,772)				UNITED STATES: NLSY (N=1,577)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Adult height (inches)	.027 (.0053)	.022 (.0052)	.004 (.0074)	.005 (.0073)	.025 (.0062)	.018 (.0060)	.002 (.0096)	-.004 (.0091)
Youth height (inches)			.026 (.0066)	.021 (.0066)			.027 (.0095)	.026 (.0090)
Age	...	...	...	...	.028 (.0066)	.027 (.0065)	.024 (.0067)	.023 (.0065)
Mother's years of schooling		.016 (.0104)		.016 (.0104)		.025 (.0092)		.023 (.0092)
Mother skilled/professional		-.080 (.0357)		-.074 (.0356)		.019 (.0608)		.024 (.0606)
Father's years of schooling		.008 (.0086)		.007 (.0087)		.030 (.0065)		.030 (.0065)
Father skilled/professional		.135 (.0467)		.130 (.0465)		.050 (.0459)		.052 (.0458)
Number of siblings		-.033 (.0084)		-.029 (.0084)		-.023 (.0077)		-.023 (.0077)
Adjusted R^2	.032	.047	.037	.049	.031	.092	.034	.094
F-statistic ($K, N - K - 1$)	9.99	10.25	11.47	10.97	9.86	15.52	8.82	14.31

$$Y = \beta_0 + \beta_1 X + u$$

$$Y = \beta_0 + \alpha_1 X + \alpha_2 M + u$$

NOTE.—Standard errors robust to heteroskedasticity are in parentheses. See the note to table 2. The sample consists only of white male, full-time workers. Each specification includes controls for region and a constant term (results omitted).

5. Real examples

(Ye et al., 2023, JAE): Y on X, moderated by M

- A natural experiment +: management learning from stock price
- Learning from stock price: the sensitivity of firm investment on firm Tobin Q
- X: $Q \times \text{shock}$; DiD
- Y: investment

Mechanism: If the mechanism is management learning from price, then the learning should be + if

- The firm price should have information new to managers or outside managers control:
 - Growth opportunities: growth companies (M/B ratio)
 - Macroeconomic information: earnings' co-movement with GDP or energy price
 - Can be continuous M or grouped dummy (1 if $M > \text{median } M$, 0 otherwise, 2, 3, 4,5, 10 groups)

5. Real examples: (Ye et al., 2023, JAE): Y on X, moderated by M

Panel A: Effects of the Increase in the Tick Size on Investment–*q* Sensitivity

	<i>INV</i>	
	(1)	(2)
<i>Q</i>	1.57*** (0.16)	1.61*** (0.16)
<i>TREAT</i> × <i>POST</i>	−1.84*** (0.61)	−1.43** (0.62)
<i>TREAT</i> × <i>Q</i>	−0.23 (0.26)	−0.19 (0.25)
<i>POST</i> × <i>Q</i>	−0.05 (0.19)	0.02 (0.19)
<i>TREAT</i> × <i>POST</i> × <i>Q</i>	0.96*** (0.30)	0.82*** (0.29)

$Y = \beta_0 + \beta_1 X + u$

	$Y = \beta_0 +$		<i>INV</i>	
	(1)	(2)	(1)	(2)
	<i>INV</i>	Macroeconomic exposure proxy	<i>CYCLICALITY</i>	<i>ENERGY</i>
<i>TREAT_GROWTH</i> × <i>POST</i> × <i>Q</i>	0.95*** (0.24)	<i>TREAT_HIGHMACRO</i> × <i>POST</i> × <i>Q</i>	1.28*** (0.41)	1.19*** (0.37)
<i>TREAT_VALUE</i> × <i>POST</i> × <i>Q</i>	0.41 (0.67)	<i>TREAT_LOWMACRO</i> × <i>POST</i> × <i>Q</i>	0.05 (0.71)	0.32 (0.56)
Controls	Y	Controls	Y	Y
Firm & Year FE	Y	Firm & Year FE	Y	Y
Obs.	5544	Obs.	5542	5542
Adj. <i>R</i> ²	0.80	Adj. <i>R</i> ²	0.80	0.80

5. Real examples

(Di Giuli and Laux, 2022, JFE): X as IV of M to Y

- Media-link on firm's financing decision
- X: media-linked director (co-director with media companies)
- Y: Firm's financing: + bond, - bank loan, - blockholder ownership%

Mechanism: media-linked director => + news coverage (M)

- Media as external governance (Dyck et al., 2008, JF)
- => - other governance necessity (bank & blockholder)

5. Real examples: (Ye et al., 2023, JAE): Y on X, moderated by M

Regressors (One lag of:)	Financing change variable			
	(1)	(2)	(3)	(4)
	<i>Public debt issued</i>	<i>New secured bank loans</i>	<i>Equity blockholder change</i>	<i>New equity issued</i>
<i>MLD</i>	0.103 (2.24)	-0.114 (-2.09)	-0.120 (-2.90)	-0.076 (-1.46)

$$Y = \beta_0 + \beta_1 X + u$$

	Fin. change variable			
	(1)	(2)	(3)	(4)
	<i>Public debt issued</i>	<i>New secured bank loans</i>	<i>Equity blockholder change</i>	<i>New equity issued</i>
Panel A: Coefficient estimates with actual lagged <i>MLD</i> as news driver				
Subpanel A1: <i>Log news count</i> as news outcome measure				
Effect of <i>MLD</i> on <i>Log news count</i>	0.098 (6.67)	0.098 (6.67)	0.098 (4.18)	0.098 (6.67)
Effect of <i>Log news count</i> on <i>Fin. change</i>	0.521 (1.88)	-0.558 (-2.10)	-0.058 (-2.01)	-0.394 (-1.11)

$$M = \beta_0 + \alpha_1 X + u$$

$$Y = \beta_0 + \gamma_1 \hat{M} + u$$